Cinematic Sentiment AI

End-to-End Movie Review Analysis

From ML Baselines to LLM Agents

Team

· Youssef Mostafa

· Omar Diaa

· Rofaida Samy

· Mohab Hossam

· Abo Bakr Zakaria

Project Highlights

96.88%

RoBERTa Accuracy

9

Models Compared

50K

IMDB Reviews

The Problem & Dataset

The Goal

Build a system that reads movie reviews and accurately classifies them as Positive or Negative sentiment.

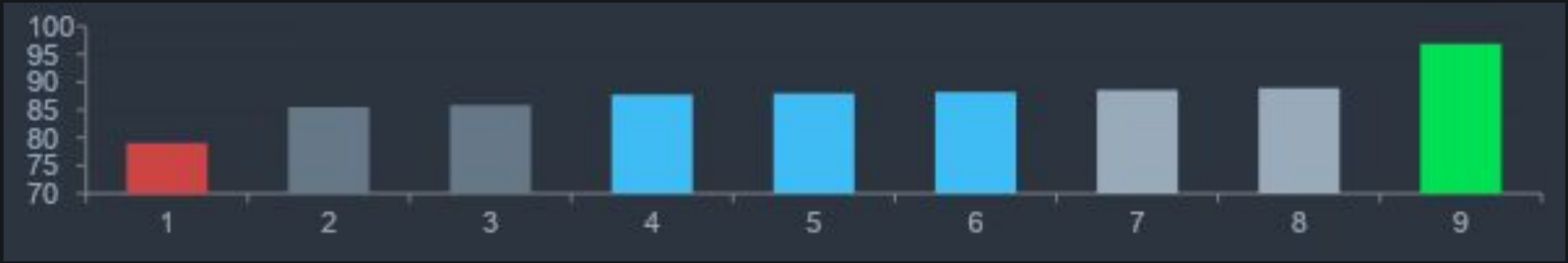
The Dataset

IMDB 50K Movie Reviews — 25,000 positive + 25,000 negative. Perfectly balanced, the gold-standard NLP benchmark.

The Challenge

Reviews are long (200+ words), nuanced, and full of sarcasm. We need context, not just keyword matching.

Accuracy Journey — All 9 Models (Post Bug-Fix Results)



Phase 1 — Preprocessing & The Negation Bug

Lowercase — Strip HTML — Lemmatize — Filter Stops — Clean Text

Standard Cleaning

- Lowercase all text for uniform tokenization
- Remove
 HTML tags from IMDB raw data
- NLTK WordNetLemmatizer — reduce inflected forms
- Keras Tokenizer — vocab size 20,000

⚠ The Negation Bug

- Standard stopwords removed "not", "never", "no"
- "This movie is not good" → "movie good" ✗
- Sentiment completely flipped by removal
- Fix: custom negation-aware filter preserves these

✓ Our Solution — Negation-Aware Filter

Explicitly subtracted "not", "no", "nor", "never", "neither", "without" from the NLTK stopword list. GloVe pre-training also encodes these words with meaningful directional semantics — retaining them benefits both the embedding layer and the tokenizer vocabulary.

Phase 2 — Machine Learning Baselines

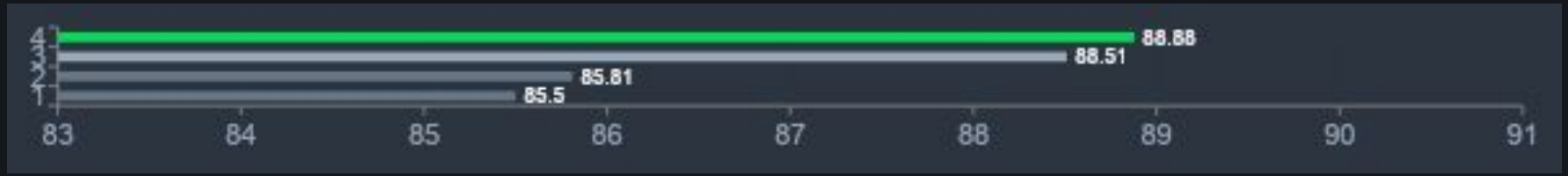
Feature Engineering

- TF-IDF: Unigrams + Bigrams (max 5,000 features)
- Bigrams capture phrases like "waste of time"
- Extra features: word count, char count, punct density
- Balanced classes — no SMOTE needed (50/50 split)

Logistic Regression	88.88%
Linear SVM	88.51%
Random Forest	85.5%
Naive Bayes	85.81%

Key Finding — Why ML slightly beat DL

TF-IDF excels at raw keyword spotting — words like "terrible" or "masterpiece" are direct, unambiguous signals. Logistic Regression (88.88%) edged out GRU/Bi-LSTM (~88%) because LSTMs require vastly more data to learn sequence context without overfitting. On 50K reviews, TF-IDF's keyword advantage is simply hard to beat.



Phase 3 — Deep Learning: The Bug & The Fix

The Keras Callback Bug — Why GRU/Bi-LSTM appeared to fail at ~50%

EarlyStopping + ReduceLROnPlateau callbacks shared internal state across models in our training for-loop, killing GRU and Bi-LSTM prematurely on Epoch 3 before gradients converged. Fix: separate callback instances per model. Result: ~50% → ~88%.

Vanilla RNN

79.0%

VANISHING GRAD

No memory gates — gradient vanishes over 200-word sequences. Expected architectural limit.

LSTM

87.7%

WORKING ✓

Stable after callback fix. Long-range dependencies captured via input/forget/output gates.

GRU

88.2%

BEST DL ✓

Fastest convergence. Matches Bi-LSTM accuracy with fewer parameters — efficient architecture.

Bi-LSTM

88.0%

WORKING ✓

Bidirectional reading adds context. Nearly identical to GRU — both solve vanishing gradients.

Vanilla RNN — Still fails (79%)

- No gating mechanism — cannot selectively remember
- Gradient vanishes over 200-word sequences regardless
- Not a bug — this is the expected architectural limit

LSTM / GRU / Bi-LSTM — Solved (88%)

- Memory gates successfully prevent vanishing gradients
- GRU and Bi-LSTM now match ML baselines (~88%)
- The bug was the culprit — architecture was always right

Phase 4 — The Transformer Pivot (SOTA)

96.88%

RoBERTa Accuracy

0.97 Precision · 0.97 Recall

+8.0% vs Logistic Regression

160GB Pre-training

Trained on 10× more text than BERT — Wikipedia, books, web pages. Deep English understanding baked in before fine-tuning.

Self-Attention

Reads all 512 tokens at once. Combines keyword spotting WITH full context — no forgotten beginnings, no gradient issues.

Why it wins by 8%

ML spots keywords. DL reads sequences. RoBERTa does BOTH simultaneously. That dual capability is the 8% gap over everything else.

Full Model Leaderboard

Model	Type	Accuracy	vs Log Reg
Vanilla RNN	DL	79.04%	−9.84%
Random Forest	ML	85.50%	−3.38%
LSTM	DL	87.72%	−1.16%
Bi-LSTM	DL	88.01%	−0.87%
GRU	DL	88.19%	−0.69%
Linear SVM	ML	88.51%	−0.37%
Logistic Regression	ML	88.88%	baseline
RoBERTa ★	**Transformer**	**96.88%**	**+8.00%**

Phase 5 — RAG & LLM Explanations

The Black Box Problem: RoBERTa outputs "Positive" with 97% confidence — but gives zero reasoning. Users need to know WHY.

User Review
Raw input text

→

FAISS Search
5,000 indexed reviews

→

Top 3 Similar
Retrieved context

→

Groq LLaMA 3
LLM generates explanation

→

Explanation
3-sentence human output

FAISS Vector Database

- 5,000 IMDB reviews indexed as 384-dim dense vectors
- Embedding: sentence-transformers/all-MiniLM-L6-v2
- Cosine similarity retrieves top 3 semantically similar
- Retrieved reviews injected as LLM prompt context

Groq LLaMA 3 Integration

- Groq API (free tier) — ultra-fast LLaMA 3 inference
- Prompt: prediction + confidence + 3 similar reviews
- Output: 3 sentences quoting key sentiment phrases
- Black-box ⟶ explainable, auditable reasoning

Example: "This review is POSITIVE (96.9%). The phrases 'masterpiece' and 'haunts you' signal strong emotional admiration. 3 similar reviews also praise cinematography and score — reinforcing this prediction."

Phase 6 — The Autonomous AI Agent

AI Agent built with Groq native function calling — users ask natural language questions; the agent autonomously triggers the correct Python tools.

classify_review()

Passes input through fine-tuned RoBERTa — returns sentiment label + confidence score.

find_similar()

Queries FAISS vector DB — retrieves top 3 semantically similar IMDB reviews as context.

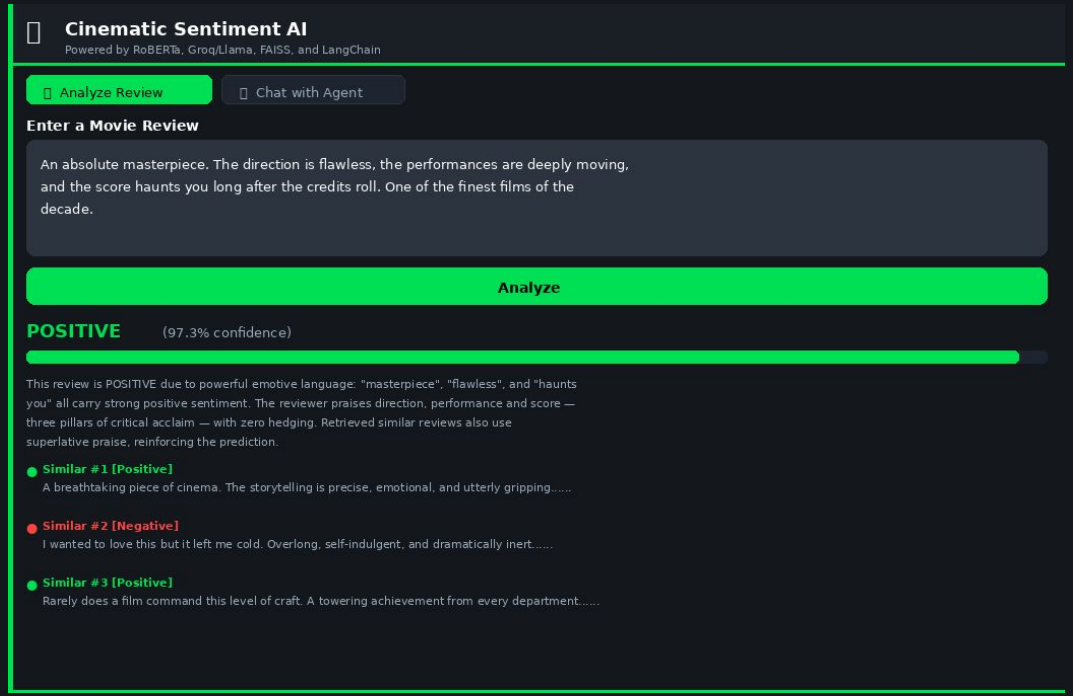
explain_sentiment()

Calls Groq LLaMA 3 with prediction + context reviews — returns a 3-sentence explanation quoting key phrases from the input review.

How It Works

User: "Analyze this review and find similar ones."
Agent: classifies → retrieves similar → explains → returns a single unified natural language response.

Phase 7 — The Final Application



Gradio UI — Interactive Demo

Tab 1 — Analysis

- Paste any review → instant RoBERTa prediction
- Confidence score with sentiment label
- LLM explanation quoting key phrases
- Top 3 similar reviews from FAISS

Tab 2 — AI Agent

- Full chat interface with Groq LLaMA 3
- Natural language queries to the agent
- Agent chains classify → retrieve → explain
- Transparent tool call logging

Stack: Gradio · HuggingFace · Groq · FAISS · sentence-transformers

Conclusion & Future Work



A Keras callback bug caused false ~50% results. After the fix, LSTM/GRU/Bi-LSTM all reached ~88% — proving gated architectures successfully solve vanishing gradients.



TF-IDF + Logistic Regression (88.88%) barely edges DL models. Keyword spotting is powerful — LSTMs need more data to leverage sequence context without overfitting.



RoBERTa (96.88%) dominates by combining self-attention with 160GB of pre-training — keyword spotting AND context simultaneously. An 8% gap over everything else.



LLM + RAG + Agent transforms a black-box classifier into an explainable, conversational AI system accessible to non-technical users.

Future Work

- Generalization test on Letterboxd 90K dataset
- Cross-domain eval on Metacritic 10K reviews
- DeBERTa-v3 fine-tuning — targeting 97%+ accuracy
- Sentiment regression (predict star rating, not binary)
- Deploy as a public web app with persistent history

Final Score

96.88%

RoBERTa · IMDB 50K · 9 Models Compared